

# DAA – Quick Practice Notes

## 1. What is an Algorithm?

An algorithm is a finite sequence of clear steps used to solve a problem or perform a task.

## 2. Characteristics of an Algorithm

- Input – accepts zero or more inputs.
- Output – produces at least one result.
- Definiteness – every step is clear and unambiguous.
- Finiteness – terminates after a finite number of steps.
- Effectiveness – each step can be carried out in a practical way.

## 3. Time Complexity

Time complexity describes how the running time of an algorithm grows as the input size increases.

| Complexity  | Example                       |
|-------------|-------------------------------|
| $O(1)$      | Accessing an array element    |
| $O(n)$      | Linear search                 |
| $O(\log n)$ | Binary search                 |
| $O(n^2)$    | Simple nested-loop comparison |

## 4. Mini Example – Linear Search

Given the list [10, 25, 30, 45] and target 30, compare the target with each element from left to right. The target is found at the third position. In the worst case, linear search takes  $O(n)$  time.

## Practice Questions

1. Define an algorithm in your own words.
2. What is the difference between  $O(1)$  and  $O(n)$ ?
3. Give one real-life example of an algorithm.
4. What is the worst-case time complexity of linear search?